

VINH NGUYEN

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EDUCATION

University of Minnesota - Twin Cities, Minneapolis, MN

Master of Science (M.S.) in Computer Science

Expected December 2025

Bachelor of Science (B.S.) in Computer Science

December 2024

- GPA: 3.8/4.0

Relevant Coursework: Machine Learning Fundamentals, Data Mining, Advanced Machine Learning, Deep Learning, AI-enabled Visual MR System, Secure Software Systems Development

WORK EXPERIENCE

Research Assistant

September 2024 - Present

University of Minnesota, College of Science and Engineering

- Meet weekly with the professor to discuss task and research direction
- Meet weekly with a team from Harvard's Visual Computing Group to collaborate in developing a LLM-based video editing and analysis tool
- Develop and fine-tune computer vision models for better segmentation, pose estimation, and object detection
- Conduct user studies on data visualization and the effect of augmenting videos with data

Software Development Intern

January 2023 - May 2024

Hennepin County IT Department

- Developed full-stack custom web applications using Angular and .NET frameworks
- Worked closely with senior developers and other team members to implement new features during development iterations
- Gained extensive experience in both user interfaces and API development, as well as knowledge in software development life cycle
- Gained experience working with Microsoft Azure for cloud hosting of applications' databases

Embedded Systems Developer, Command and Data Handling

September 2021 - April 2022

Small Satellite Research Lab at UM-TC

- Coded, tested, and debugged communication protocols such as SPI and I2C
 - Attended weekly meetings with other team members to collaborate and discuss new tasks
 - Programmed Raspberry Pi's and microcontrollers in C++ to interact with other hardware components
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PROJECT

MatrixForge - A Linear Algebra and Machine Learning Library in C

github.com/NguyenLTVinh/MatrixForge

- Built a multithreaded C library for neural network training and matrix operations
- Optimized performance using memory alignment and OpenMP parallelism
- Achieved handwritten digit classification using Leaky ReLU and gradient descent

Two-Factor Authenticated Door Lock System

github.com/NguyenLTVinh/2FA-Lock-System

- Developed a secure RFID and PIN-based door lock system with Bluetooth communication
 - Integrated AVR microcontrollers, cryptography chips, and LCD/keypad interfaces
 - Enforced reliability with modular design and anti-forgery mechanisms
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SKILLS

Programming languages: Python, C, C++, Rust, C#, JavaScript, TypeScript, Java

Frameworks: TensorFlow, PyTorch, OpenCV, React.js, Angular, .NET, Express

Other tools: Linux (Debian based), SQL, Git, MATLAB, Anaconda, Python Machine Learning libraries

Other skills: Identifying and solving problems, data analysis, writing, teamwork, organization